

Smartphones made of deep-sea rocks

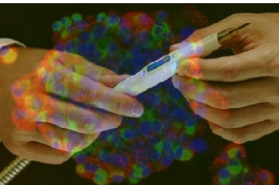
In a bid to address the issue of depleting materials used in the manufacture of smartphones, geochemistry researchers at Jacobs University in Germany are currently working on a new ore-leaching method to extract rare earth metals buried in deep sea/ocean floors. Further, the same rare earth metals can be used as viable substitutes to materials of usual smartphone parts.

It is a wide-known fact that the deep sea/ocean floor is a rich reservoir of iron and manganese also known as ferromanganese deposits. The researchers have devised an efficient way to extract these from ferromanganese nodules located in the ocean floor. A new ore-leaching method has made it possible to extract up to 80 per cent of four rare earth metals from some such nodules.

Once extracted, the metals can then be used for a wide range of applications ranging from solar panels, wind turbines to well of course your next smartphone.

New smart biosensor that directs cells to kill cancer

Biologists at the Northwestern University's McCormick School of Engineering and Applied Science have developed a ground-breaking technology that could modify human cells to create therapeutics used in turn to selectively target and destroy tumour cells in the human body without disrupting healthy cells. The unique protein biosensor engineers cells to kill cancer by helping them effectively distinguish between healthy and cancerous cells.



Smart device to kill cancer

While sitting on the surface of a cell, the biosensor can be programmed to sense its immediate environment for specific factors following which it sends a signal to the engineered cell's nucleus. This triggers a gene expression programme within the cell.

"Till date, there was no way to engineer cells in a manner that allowed them to sense key pieces of information about their environment, which could indicate whether the engineered cell is in healthy tissue or sitting next to a tumour," Joshua Leonard, an assistant professor at Northwestern University's McCormick School of Engineering and Applied Science was quoted as saying.

Charge your smartphone in mere 30 seconds

Honestly, smartphone battery life remains a big frustration amongst us. However, if an Israeli startup called



StoreDot batteries charge in just 30 seconds

StoreDot has its way, we might soon be able to rid ourselves of the tedious task that is charging, well not entirely! StoreDot's prototype charger can replenish your smartphone battery in half a minute. However, there is a catch.

StoreDot recently demonstrated their prototype charger at Microsoft's Think Next Conference successfully charging a Samsung Galaxy S4's battery (2600mAh) in under half a minute. The catch is the charger only works with the Galaxy S4 for now; however, the company is working to get their charger work with all handsets.

Ironically, a magic charger was not on their mind when StoreDot started working on developing 'Nanodots' comprising bio-organic peptide molecules. Interestingly, these peptide

molecules are set into random motion once connected to a smartphone battery, substantially reducing charging time in the process.

StoreDot's unique charger will be a long time coming as far as commercial availability is concerned: post 2016 to be precise. Currently, the developers are working to reduce the size of the charger that at present is of the size of an average laptop charger.

Disposable mobile phones with throwaway processors

In a recent development, researchers at University of Iowa have proposed a new method to pave the way for flexible plastics that could substitute the traditional silicon chips in computers, mobile phones and other systems.

Michael Flatte, a professor of physics and astronomy at the College of Liberal Arts and Sciences and his colleagues have devised a method that could overcome the requirement of large amount of energy currently needed to read stored information in such plastics. The research opens the door for inexpensive disposable mobile devices in the future.

While encoding information is usually preferred in light for fibre-optic transmission, magnetism is a much preferred choice when storing information to ensure it survives for years with no additional power to maintain the same.

While converting energy from one form to another is cost-effective in case of silicon-chip-based computers, flexible plastic computing devices call for greater costs therefore deterring their use, until now. The researchers have succeeded in transducing information between a magnet and an organic light-emitting diode at room temperature and without electrical current flow.

A critical issue is how to convert information from one type to another. Here, Flatte and his colleagues have shown an efficient means of converting information encoded in magnetic storage to light in a flexible plastic device. The magnetic fields from the magnetic storage device directly modify the light emission from the device.

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